INSTITUTO COLLEGE

MATH 110: General Education Math

Fall 2026

Sets Review Solutions

Sets and Elements

A **set** is a collection of distinct objects, called its **elements**. We list the elements of a set between braces, for example $A = \{\text{yellow}, -4, \text{raincoat}\}$ denotes a set with 3 elements – the elements are yellow, -4, and raincoat. We write $x \in A$ when x is an element of A, and $x \notin A$ when it is not. The order of the elements does not matter, and each element is listed only once. The **empty set**, denoted $\emptyset$, is the unique set that contains no elements.

1. Let $R = \{\text{red}, \text{green}, \text{blue}\}$ and $N = \{2, 7, 9\}$. Write *true* or *false*.

   (a) green $\in R$ true

   (b) yellow $\in R$ false

   (c) yellow $\notin R$ true

   (d) $7 \in N$ true

   (e) $3 \in N$ false

   (f) red $\in N$ false

   (g) $\{2, 7, 9\} = \{9, 2, 7\}$ true

2. Write each set by listing its elements between braces.

   (a) W, the vowels in the word NURSE. $W =$ $\{\text{U}, \text{E}\}$

   (b) T, the even numbers between 1 and 9. $T =$ $\{2, 4, 6, 8\}$

   (c) J, the months of the year whose names begin with the letter J.
   $J =$ $\{\text{January}, \text{June}, \text{July}\}$

   (d) Z, the whole numbers that are greater than 5 and less than 6.
   $Z =$ $\emptyset$ (there are none)

Cardinality

The **cardinality** of a set A, written $|A|$, is the number of elements in A. A set is a collection of objects, but its cardinality is a number. The empty set, $\emptyset$, is the unique set with no elements, so $|\emptyset| = 0$.

3. Find the cardinality of each set.

   (a) $\big|\{\text{red}, \text{green}, \text{blue}\}\big| =$ 3

   (b) $\big|\{2, 7, 9\}\big| =$ 3

   (c) $\big|\{100\}\big| =$ 1

   (d) $|\emptyset| =$ 0

   (e) $\big|\{0\}\big| =$ 1

   (f) $\big|\{\text{Mon}, \text{Wed}\}\big| =$ 2

In part (e), the set $\{0\}$ has one element, the number 0. It is not the empty set.

4. Let $N = \{2, 7, 9\}$.

   (a) Find $|N|$. $|N| =$ 3

   (b) Is $|N|$ an element of N? Explain.

   Answer: No. $|N| = 3$, and $3 \notin N$. The elements of N are 2, 7, and 9. The cardinality 3 is not one of them: it counts them.

   (c) A student writes $N = 3$. What is wrong with this? What should the student write instead?

   Answer: N is a set and 3 is a number, so they cannot be equal. The student should write $|N| = 3$.

5. Each set below is described in words. Find its cardinality without listing all of its elements.

(a) E, the set of even numbers between 1 and 100, including 100. $|E| =$ 50

Answer: The elements of E are 2, 4, 6, ..., 100. Each one is 2 times one of the numbers $1, 2, 3, \ldots, 50$, so E has 50 elements.

(b) W, the set of whole numbers from 1 to 250. $|W| =$ 250

Answer: The elements of W are 1, 2, 3, ..., 250, so W has 250 elements.

(c) T, the set of whole numbers from 10 to 30. $|T| =$ 21

Answer: The whole numbers from 1 to 30 are 30 numbers. Leaving out 1 through 9 leaves $30 - 9 = 21$. (Not $30 - 10 = 20$: both 10 and 30 are elements of T.)

(d) M, the set of multiples of 5 from 5 to 100. $|M| =$ 20

Answer: The elements of M are 5, 10, 15, ..., 100. Each one is 5 times one of the numbers $1, 2, 3, \ldots, 20$, so M has 20 elements.

(e) O, the set of odd numbers between 0 and 20. $|O| =$ 10

Answer: The elements of O are 1, 3, 5, ..., 19. Half of the 20 whole numbers from 1 to 20 are odd, so O has 10 elements.

(f) D, the set of days in the month of November. $|D| =$ 30

Answer: November has 30 days.

Set or Number?

6. For each expression, write *set* or *number*.

Expression	Set or number?
$\{4\}$	set
4	number
$\lvert\{4\}\rvert$	number
$\emptyset$	set
$\lvert\emptyset\rvert$	number
$\{\text{Ana}, \text{Ben}\}$	set
$\lvert\{\text{Ana}, \text{Ben}\}\rvert$	number
0	number

Union and Intersection

The **union** of two sets A and B, denoted $A \cup B$, is the set of elements that are in A or in B (or in both). The **intersection** of A and B, denoted $A \cap B$, is the set of elements that are in both A and B. The union and the intersection of two sets are sets, so when we write their elements, we use braces. Often all of the sets we are working with are made up of elements from one larger set, called the **universal set** U, which consists of all the elements we are working with.

The clinic. Many of the problems below are about one clinic. Eight patients visit the clinic today: Ana, Ben, Carla, David, Eva, Hugo, Iris, and Juan. Using their first initials, the universal set is

$$U = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{H}, \text{I}, \text{J}\}.$$

The patients who have a fever form the set $F = \{\text{A}, \text{C}, \text{D}, \text{I}\}$, and the patients who have had a flu shot this year form the set $V = \{\text{B}, \text{C}, \text{E}, \text{I}, \text{J}\}$.

7. Use the clinic sets U, F, and V.

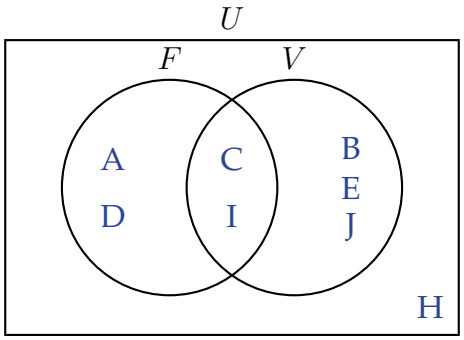
   (a) Find $F \cup V$. $F \cup V =$ $\{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{I}, \text{J}\}$

   (b) Find $F \cap V$. $F \cap V =$ $\{\text{C}, \text{I}\}$

   (c) Find each cardinality.

   $|F| =$ 4 $\quad |V| =$ 5 $\quad |F \cup V| =$ 7 $\quad |F \cap V| =$ 2

8. Use the clinic sets U, F, and V. Write the initial of each of the 8 patients in the correct region of the Venn diagram.



   (a) Which patients are in F but not in V? Ana and David

   (b) Which patients are in both F and V? Carla and Iris

   (c) Which patients are in neither F nor V? Hugo

Subsets

A set A is a **subset** of a set B, written $A \subseteq B$, if every element of A is also an element of B. Every set is a subset of itself, so $A \subseteq A$, and the empty set is a subset of every set, so $\emptyset \subseteq A$.

9. Use the clinic sets U, F, and V, and your answers to Problems 7 and 8. Write *true* or *false*. If a statement is false, name an element that shows it is false.

(a) $F \subseteq F \cup V$ true

(b) $V \subseteq F \cup V$ true

(c) $F \cap V \subseteq F$ true

(d) $F \cap V \subseteq V$ true

(e) $F \subseteq V$ false

(f) $F \cup V \subseteq F$ false

(g) $F \subseteq F \cap V$ false

(h) $F \subseteq U$ true

In part (e), Ana is in F but not in V. In part (f), Ben is in $F \cup V$ but not in F. In part (g), Ana is in F but not in $F \cap V$.

10. Let $S = \{1, 2, 3\}$.

(a) List every subset of S in the table, sorted by the number of elements.

Number of elements	Subsets of S	How many?
0	$\emptyset$	1
1	$\{1\}$, $\{2\}$, $\{3\}$	3
2	$\{1, 2\}$, $\{1, 3\}$, $\{2, 3\}$	3
3	$\{1, 2, 3\}$	1

(b) How many subsets does S have in all?

Answer: $1 + 3 + 3 + 1 = 8$

(c) Explain why $\emptyset$ and S itself are both subsets of S.

Answer: Every set is a subset of itself, so $S \subseteq S$. The empty set has no elements, so it has no element that fails to be in S, and $\emptyset \subseteq S$.

The symbols $\in$ and $\subseteq$ mean different things. In $x \in A$, the thing on the left is an *element*: $x \in A$ says that x is one of the elements of A. In $A \subseteq B$, the thing on the left is a *set*: $A \subseteq B$ says that every element of A is also an element of B. The symbol '$\in$' is read "is a member of" or "is in" and the symbol '$\subseteq$' is read "is a subset of" or "is contained in."

11. In parts (a)–(j), let $N = \{2, 7, 9\}$. In parts (k)–(p), use the clinic sets U, F, and V.

For parts (a)–(j), write *true* or *false*.

(a) $7 \in N$ ___true___

(b) $\{7\} \subseteq N$ ___true___

(c) $\{7\} \in N$ ___false___

(d) $7 \subseteq N$ ___false___

(e) $\{2, 9\} \subseteq N$ ___true___

(f) $\{2, 9\} \in N$ ___false___

(g) $\emptyset \subseteq N$ ___true___

(h) $\emptyset \in N$ ___false___

(i) $N \subseteq N$ ___true___

(j) $N \in N$ ___false___

The elements of N are the numbers 2, 7, and 9. The sets $\{7\}$, $\{2, 9\}$, $\emptyset$, and N are not elements of N, so (c), (f), (h), and (j) are false. In (d), 7 is a number, not a set, so it cannot be a subset of anything.

For parts (k)–(p), fill in each blank with $\in$ or $\subseteq$.

(k) C ___$\in$___ F

(l) {C} ___$\subseteq$___ F

(m) {C, I} ___$\subseteq$___ V

(n) I ___$\in$___ $F \cap V$

(o) $F \cap V$ ___$\subseteq$___ F

(p) H ___$\in$___ U

12. Let A and B be any two sets.

(a) Explain why $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

Answer: Every element of A is in A or in B, so it is an element of $A \cup B$. Similarly, every element of B is an element of $A \cup B$.

(b) Explain why $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

Answer: Every element of $A \cap B$ is in both A and B. In particular, it is an element of A, and it is an element of B.

(c) In a Venn diagram, how can you see that $A \cap B \subseteq A$ and $A \subseteq A \cup B$?

Answer: The region for $A \cap B$, where the circles overlap, lies inside circle A, and circle A lies inside the region for $A \cup B$, which is both circles together.

(d) Explain why $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

Answer: The empty set has no elements, so the elements that are in A or in $\emptyset$ are just the elements of A: $A \cup \emptyset = A$. No element is in both A and $\emptyset$, since nothing is in $\emptyset$, so $A \cap \emptyset = \emptyset$.

Complements

If A is a subset of a universal set U, then the **complement** of A, denoted A^c, is the set of elements of U that are not in A. The complement of a set is a set, so when we write its elements, we use braces.

13. Use the clinic sets U, F, and V. Find each set.

(a) $F^c =$ $\{B, E, H, J\}$

(b) $V^c =$ $\{A, D, H\}$

(c) $(F \cup V)^c =$ $\{H\}$

(d) $(F \cap V)^c =$ $\{A, B, D, E, H, J\}$

(e) $F \cap V^c =$ $\{A, D\}$

(f) Describe the sets in parts (c) and (e) in words.

Answer: $(F \cup V)^c$ is the set of patients who do not have a fever and have not had a flu shot. $F \cap V^c$ is the set of patients who have a fever but have not had a flu shot.

14. Use the clinic sets U, F, and V.

(a) Find U^c.

Answer: Every patient is in U, so no element of U is outside U: $U^c = \emptyset$.

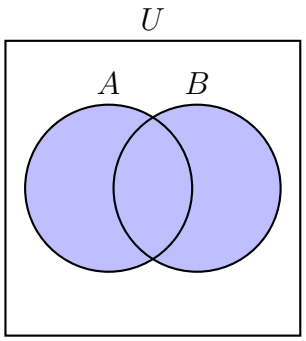
(b) Find $\emptyset^c$.

Answer: No patient is in $\emptyset$, so every element of U is in $\emptyset^c$. That is, $\emptyset^c = U$.

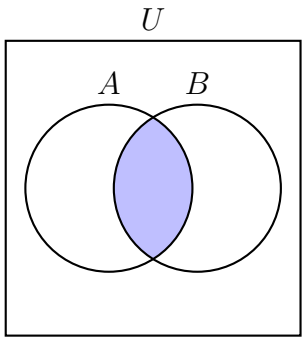
(c) Now let U be any universal set. Explain why $U^c = \emptyset$ and $\emptyset^c = U$.

Answer: U^c is the set of elements of U that are not in U. There are none, so $U^c = \emptyset$. $\emptyset^c$ is the set of elements of U that are not in $\emptyset$. Since $\emptyset$ has no elements, that is every element of U, so $\emptyset^c = U$.

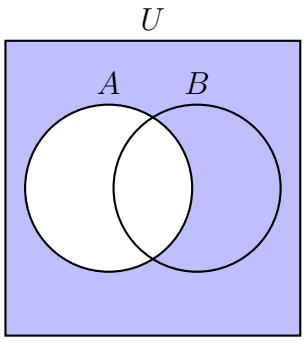
15. In each Venn diagram, the square represents a universal set U, and the circles represent subsets A and B of U. Shade the region that represents the given set.



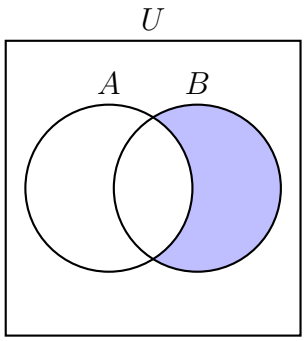
(a) $A \cup B$



(b) $A \cap B$



(c) A^c



(d) $B \cap A^c$

Set or Number? Round Two

16. Use the clinic sets U, F, and V. For each expression, write *set* or *number*, and then find it.

Expression	Set or number?	What is it?
$F \cap V$	set	$\{\text{C}, \text{I}\}$
$\lvert F \cap V \rvert$	number	2
F^{c}	set	$\{\text{B}, \text{E}, \text{H}, \text{J}\}$
$\lvert U \rvert - \lvert F \rvert$	number	$8 - 4 = 4$
$\lvert F \cup V \rvert$	number	7
U	set	$\{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{H}, \text{I}, \text{J}\}$
$\lvert U \rvert$	number	8
the patients who have a fever	set	$F = \{\text{A}, \text{C}, \text{D}, \text{I}\}$
the number of patients who have a fever	number	$\lvert F \rvert = 4$

17. Use the clinic sets U, F, and V. A student gives the answers below. Explain what is wrong with each answer, and write the correct answer.

(a) $F \cap V = 2$

Answer: $F \cap V$ is a set, not a number: $F \cap V = \{\text{C}, \text{I}\}$. (The number 2 is its cardinality: $|F \cap V| = 2$.)

(b) $|F| = \{\text{A}, \text{C}, \text{D}, \text{I}\}$

Answer: $|F|$ is a number, not a set: $|F| = 4$. (The set $\{\text{A}, \text{C}, \text{D}, \text{I}\}$ is F itself.)

(c) $F \cup V = \{\text{A}, \text{C}, \text{D}, \text{I}, \text{B}, \text{C}, \text{E}, \text{I}, \text{J}\}$

Answer: Carla and Iris are listed twice. Each element is listed only once: $F \cup V = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{I}, \text{J}\}$.

(d) $F^{\text{c}} = \{\text{B}, \text{E}\}$

Answer: Hugo and Juan are not in F either, so they are also in F^{c}: $F^{\text{c}} = \{\text{B}, \text{E}, \text{H}, \text{J}\}$.

(e) $F^{\text{c}} = 4$

Answer: F^{c} is a set, not a number: $F^{\text{c}} = \{\text{B}, \text{E}, \text{H}, \text{J}\}$. (Its cardinality is $|F^{\text{c}}| = 4$.)

(f) $F \cap V = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{I}, \text{J}\}$

Answer: This is the union $F \cup V$. The intersection contains only the patients in both sets: $F \cap V = \{\text{C}, \text{I}\}$.

Disjoint Sets

Sets A and B are **disjoint** if $A \cap B = \emptyset$. In other words, disjoint sets have no elements in common: in a Venn diagram, they do not overlap.

18. Use the clinic sets U, F, and V. Ben, Eva, and Hugo go home before noon. They form the set $G = \{\text{B}, \text{E}, \text{H}\}$.

(a) Find $F \cap G$. Are F and G disjoint?

Answer: $F \cap G = \emptyset$, so F and G are disjoint.

(b) Find $V \cap G$. Are V and G disjoint?

Answer: $V \cap G = \{\text{B}, \text{E}\}$. This is not $\emptyset$, so V and G are not disjoint.

19. Use the clinic sets U, F, and V.

(a) Find $F \cap \emptyset$. Are F and $\emptyset$ disjoint?

Answer: $F \cap \emptyset = \emptyset$, so F and $\emptyset$ are disjoint.

(b) Let A be any set. Explain why A and $\emptyset$ are disjoint.

Answer: The empty set has no elements, so no element is in both A and $\emptyset$. That is, $A \cap \emptyset = \emptyset$ (as in Problem 12(d)), so A and $\emptyset$ are disjoint.

The Addition Principle

The Addition Principle: If A and B are disjoint sets, then

$$|A \cup B| = |A| + |B|.$$

Since $A \cap B = \emptyset$, no element is in both sets, so adding $|A|$ and $|B|$ counts each element of $A \cup B$ exactly once. The same is true for three or more sets, as long as no two of them have elements in common.

20. Use the clinic sets U, F, V, and G.

(a) Find $F \cup G$ by listing its elements, and find $|F \cup G|$.

Answer: $F \cup G = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{H}, \text{I}\}$, so $|F \cup G| = 7$.

(b) Find $|F| + |G|$. Explain why your answer is the same as $|F \cup G|$.

Answer: $|F| + |G| = 4 + 3 = 7$. In Problem 18(a), $F \cap G = \emptyset$, so F and G are disjoint, and the Addition Principle says $|F \cup G| = |F| + |G|$.

(c) The sets F and F^{c} are disjoint, and $F \cup F^{\text{c}} = U$. Use the Addition Principle to find $|U|$ from $|F|$ and $|F^{\text{c}}|$.

Answer: $|U| = |F \cup F^{\text{c}}| = |F| + |F^{\text{c}}| = 4 + 4 = 8$.

(d) Can you use the Addition Principle to find $|F \cup V|$? Explain.

Answer: No. $F \cap V = \{\text{C}, \text{I}\} \neq \emptyset$, so F and V are not disjoint.

21. A hospital floor has three wards. Ward 1 has 12 patients, Ward 2 has 9 patients, and Ward 3 has 6 patients. Each patient is in only one ward. Let W_1, W_2, and W_3 be the sets of patients in Wards 1, 2, and 3.

(a) How many patients are in Ward 1 or Ward 2? Write your answer using the sets W_1 and W_2.

Answer: W_1 and W_2 are disjoint, so by the Addition Principle, $|W_1 \cup W_2| = |W_1| + |W_2| = 12 + 9 = 21$.

(b) How many patients are on the floor?

Answer: No two of the three sets have elements in common, so by the Addition Principle, $|W_1 \cup W_2 \cup W_3| = 12 + 9 + 6 = 27$.

The Inclusion-Exclusion Principle

The Inclusion-Exclusion Principle: For any two sets A and B,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Adding $|A|$ and $|B|$ counts each element of $A \cap B$ twice, once in A and once in B. Subtracting $|A \cap B|$ makes up for this, so each element of $A \cup B$ is counted exactly once.

22. Use the clinic sets U, F, and V.

(a) Find $|F|$, $|V|$, and $|F \cap V|$.

$|F| =$ 4 $\quad |V| =$ 5 $\quad |F \cap V| =$ 2

(b) Find $|F| + |V|$. Which patients are counted twice?

Answer: $|F| + |V| = 4 + 5 = 9$. Carla and Iris are in both F and V, so they are counted twice.

(c) Use the Inclusion-Exclusion Principle to find $|F \cup V|$. Check your answer by counting the elements of $F \cup V$ in Problem 7(a).

Answer: $|F \cup V| = |F| + |V| - |F \cap V| = 4 + 5 - 2 = 7$. Check: $F \cup V = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}, \text{I}, \text{J}\}$ has 7 elements.

(d) Use the Inclusion-Exclusion Principle to find $|F \cup G|$, where $G = \{\text{B}, \text{E}, \text{H}\}$. Why do you get the same answer as with the Addition Principle in Problem 20?

Answer: $|F \cup G| = 4 + 3 - 0 = 7$. Since F and G are disjoint, $F \cap G = \emptyset$, so $|F \cap G| = 0$, and there is nothing to subtract.

23. In a clinic's waiting room, 14 patients are there for a flu shot, 9 are there for a blood test, and 5 are there for both. Let S be the set of patients there for a flu shot and T the set of patients there for a blood test.

(a) Write $|S|$, $|T|$, and $|S \cap T|$.

$|S| =$ 14 $\quad |T| =$ 9 $\quad |S \cap T| =$ 5

(b) How many patients are there for a flu shot or a blood test (or both)?

Answer: By the Inclusion-Exclusion Principle, $|S \cup T| = |S| + |T| - |S \cap T| = 14 + 9 - 5 = 18$.

(c) Explain why the answer to part (b) is not $14 + 9 = 23$.

Answer: The 5 patients who are there for both are counted twice in $14 + 9$, once in S and once in T.

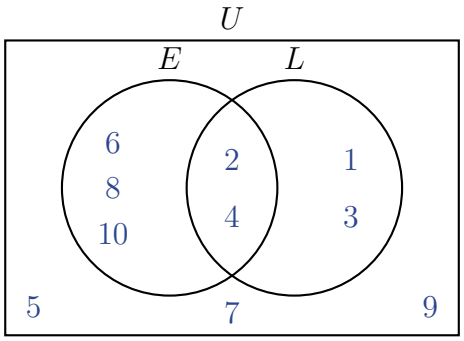
Putting It Together

24. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set. Let E be the set of even numbers in U, and let L be the set of numbers in U that are less than 5.

(a) List the elements of E and L.

$E =$ $\{2, 4, 6, 8, 10\}$ $L =$ $\{1, 2, 3, 4\}$

(b) Write each number of U in the correct region of the Venn diagram.



(c) Find each set.

$E \cup L =$ $\{1, 2, 3, 4, 6, 8, 10\}$

$E \cap L =$ $\{2, 4\}$

$E^c =$ $\{1, 3, 5, 7, 9\}$

$(E \cup L)^c =$ $\{5, 7, 9\}$

(d) Find $|E \cup L|$ by counting the elements of $E \cup L$. Then check your answer using the Inclusion-Exclusion Principle.

Answer: $E \cup L$ has 7 elements, so $|E \cup L| = 7$. Check: by the Inclusion-Exclusion Principle, $|E \cup L| = |E| + |L| - |E \cap L| = 5 + 4 - 2 = 7$.

(e) $|L| = 4$, and 4 is also an element of L. Does this mean that L and $|L|$ are the same thing? Explain.

Answer: No. $L = \{1, 2, 3, 4\}$ is a set with 4 elements, and $|L| = 4$ is a number. It is a coincidence that the number 4 is one of the elements of L.